RESHAD UL KARIM

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OBJECTIVE

A computer science researcher in machine learning – 3D vision, navigation, and language models – who adds value to a team by turning research questions into evaluated experiments and working systems, with experience as a co-author, team member, and teaching assistant. Seeking PhD opportunities to pursue this work in depth.

EDUCATION

BSc. in Computer Science and Engineering, BRAC University 09/2022 - 06/2026
CGPA: 3.89/4.00 Highest Distinction

Higher Secondary Certificate, Notre Dame College, Dhaka 06/2019 - 02/2022
GPA: 5.00/5.00

PUBLICATIONS

Journal

Optimizing Stroke Recognition with MediaPipe and Machine Learning: An Explainable AI Approach for Facial Landmark Analysis
Karim, R. U., Mahdi, S., Samin, A., Zereen, A. N., Abdullah-Al-Wadud, M. M., & Uddin, J.
IEEE Access, vol. 13, pp. 32636-32660, 2025. DOI: 10.1109/ACCESS.2025.3550577

Conference Papers

Machine Learning Approaches in Photoplethysmography-Based Sleep Stage Classification
Ferdous, T., Karim, R. U., Samin, A., Mahdi, S., Tasnim, H., & Zereen, A. N.
IEEE ICEACE 2024, Changchun, China, pp. 123-128. DOI: 10.1109/ICEACE63551.2024.10898858

Improved Photoplethysmography-Based Four-Stage Sleep Classification with Explainable AI-Driven Machine Learning
Ferdous, T., Karim, R. U., Samin, A., Mahdi, S., & Zereen, A. N.
IEEE ICEACE 2024, Changchun, China, pp. 117-122. DOI: 10.1109/ICEACE63551.2024.10898853

RESEARCH & TEACHING EXPERIENCE

Undergraduate Research Assistant 11/2025 - Ongoing
BRAC University Research Seed Grant Initiative 2025 *Dhaka, Bangladesh*
Contributing to DRISTEE, an assistive-navigation system for visually impaired users, across model integration, implementation, and academic writing.

Undergraduate Teaching Assistant 02/2025 - 06/2025
BRAC University *Dhaka, Bangladesh*
Supported 120+ students across three Numerical Methods sections through preparatory and doubt-solving sessions, one-on-one guidance, and assessment feedback.

Core Team Member – AI & Autonomous Systems 01/2024 - 06/2025
BRAC University Mars Rover Team - MONGOL TORI *Dhaka, Bangladesh*
Optimized AI-based object detection and vision-guided autonomy for the rover, and contributed to its control electronics and integration.

RESEARCH PROJECTS

Motion-Conditioned Identifiability and Abstention for Physical 3D Scene Geometry
Under double-blind review. Asks whether camera motion can settle a scene's physical explanation at all, since a real corridor, its mirror reflection, and a photo of it on a screen can produce one image. It picks the motion that best separates the closest rival explanations and abstains when none can.

Novel Positional Encoding for Long-Context Transformer LLMs
Under double-blind review. Introduces a new rotary positional encoding that helps transformer language models handle long contexts. It is evaluated by pretraining transformer models from scratch with mixed-precision training and by checking numerical stability.

Reliable Connectome-Based Autism Classification: Calibration and Explanation Stability
Under review, Computers in Biology and Medicine. Studies how reliable brain-connectivity classifiers for autism are, looking beyond accuracy to calibrated probabilities and explanations that stay stable across cohorts. It compares linear and tree-based models under controlled cohort and site-level validation.

CAMDet: Camera-Agnostic Metric Object Detection with Geometric Consistency
Under review, IEEE Transactions on Image Processing. A single-image 3D object detector that predicts object boxes and metric distance together with a per-prediction uncertainty estimate. Conditioning on camera intrinsics lets one trained model work across different depth sensors without retraining.

DRISTEE: Depth-aware Recognition and Intelligent Scene Translation for Enhanced Engagement
Submitted for journal review. An assistive navigation system for visually impaired users that combines depth-aware mapping, object and face recognition, and voice interaction to give spoken spatial guidance.

Camera-Aware Language-Guided Navigation Policy from a Single RGB Camera
In preparation. A navigation policy that follows natural-language instructions using only one ordinary camera, with no depth sensor or map. It combines vision-language models with camera-aware learning so the policy can adapt to different cameras and environments; evaluation is ongoing.

Evidence-Weighted Collaborative Gaussian-Splatting SLAM on Heterogeneous Mobile Devices
In preparation. A real-time collaborative 3D mapping system for consumer phones, where each device can map on its own and join or leave a session at any time. Observations are merged by the physical evidence they provide rather than averaged, so weaker cameras and sensors do not degrade the shared map.

LUMENAA: Multimodal Agentic AI System
Under review, Engineering Applications of Artificial Intelligence. A spoken assistant that helps blind and low-vision users understand their surroundings through a camera and voice. It judges how far to trust each of its perception tools on the current view and applies fixed safety rules, so it answers, hedges, asks for a better view, or declines when the evidence is weak.

Leveraging Paraphrase Geometry for Model-Agnostic Embedding Enhancement
Ongoing. A training-free way to improve sentence embeddings from any model. It finds the directions in embedding space that mostly encode paraphrase-related variation and removes them, which sharpens similarity and retrieval quality.

Explainable Pedestrian–Vehicle Conflict-Risk Prediction
Ongoing. Predicts pedestrian–vehicle conflict risk from dashcam video by combining road segmentation, object detection, and pose estimation, with decisions explained through feature attributions.

WeHeal: Full-Stack Telemedicine Platform
Deployed at weheal.live. A telemedicine web platform where patients, doctors, and clinic staff work in one place through role-based dashboards and video consultations, with an assistant for patient questions and scheduled automations.

Modular Vision-Based Autonomous Robot with Manipulation
Completed. A modular robot built from two networked microcontrollers that navigates between visual markers and picks up and places objects, using visual servoing to align its arm with targets seen by the camera.

Wearable Health Monitoring & Fall Detection for Elderly Care
Completed. A low-cost wearable for elderly care that combines body-sensor readings with motion sensing to detect health anomalies and falls, raising local alarms and sending SMS alerts with the wearer's location to caregivers.

HONORS & AWARDS

Highest Distinction, BRAC University
Vice Chancellor's List (5×) and Dean's List (5×), BRAC University 01/2023 - Present
Globally Ranked 8th, University Rover Challenge 2025, The Mars Society – Team BRACU Mongol-Tori
First Runner Up, System Analysis and Design Project, School of Data and Sciences, BRAC University
Second Runner Up, Robosoccer Competition TECHSPECTRA, Robotics Club of BRAC University
Third Runner Up & Grand Finalist, EXCELERATE 2025, Finance and Accounting Club of BRAC University
Semi Finalist, On-Campus Rounds, Hult Prize BRAC University 2023
Bronze Award, The Duke of Edinburgh's International Awards

RESEARCH INTERESTS & TECHNICAL SKILLS

Research Interests: 3D Vision, Trustworthy ML, Vision-Language Navigation, LLMs, Assistive AI, Biomedical AI
Computer Vision: YOLO, SegFormer, ViT/DINOv2, Monocular 3D/Metric Depth, Pose Estimation, OpenCV
ML & Deep Learning: PyTorch, Hugging Face, Scikit-learn, Transformers, Explainable AI (SHAP), Calibration
Language & Multimodal Models: Pretraining, Sentence Embeddings, Vision-Language Models
Programming & Tools: Python, C/C++, JS/TS, Swift, SQL; W&B, ONNX, RunPod, Git, Linux, ROS, ESP32